Best Time to Buy and Sell Stock II (medium)

Problem Statement

You are given a `prices` array containing the integers, where `price[i]` refers to the stock price on i^{th} day.

You are allowed to do as many transactions as you like (buy one and sell one stock multiple times). You cannot engage in multiple transactions `simultaneously` (you must sell the stock before you buy again). However, you can buy again on the `same day` after selling.

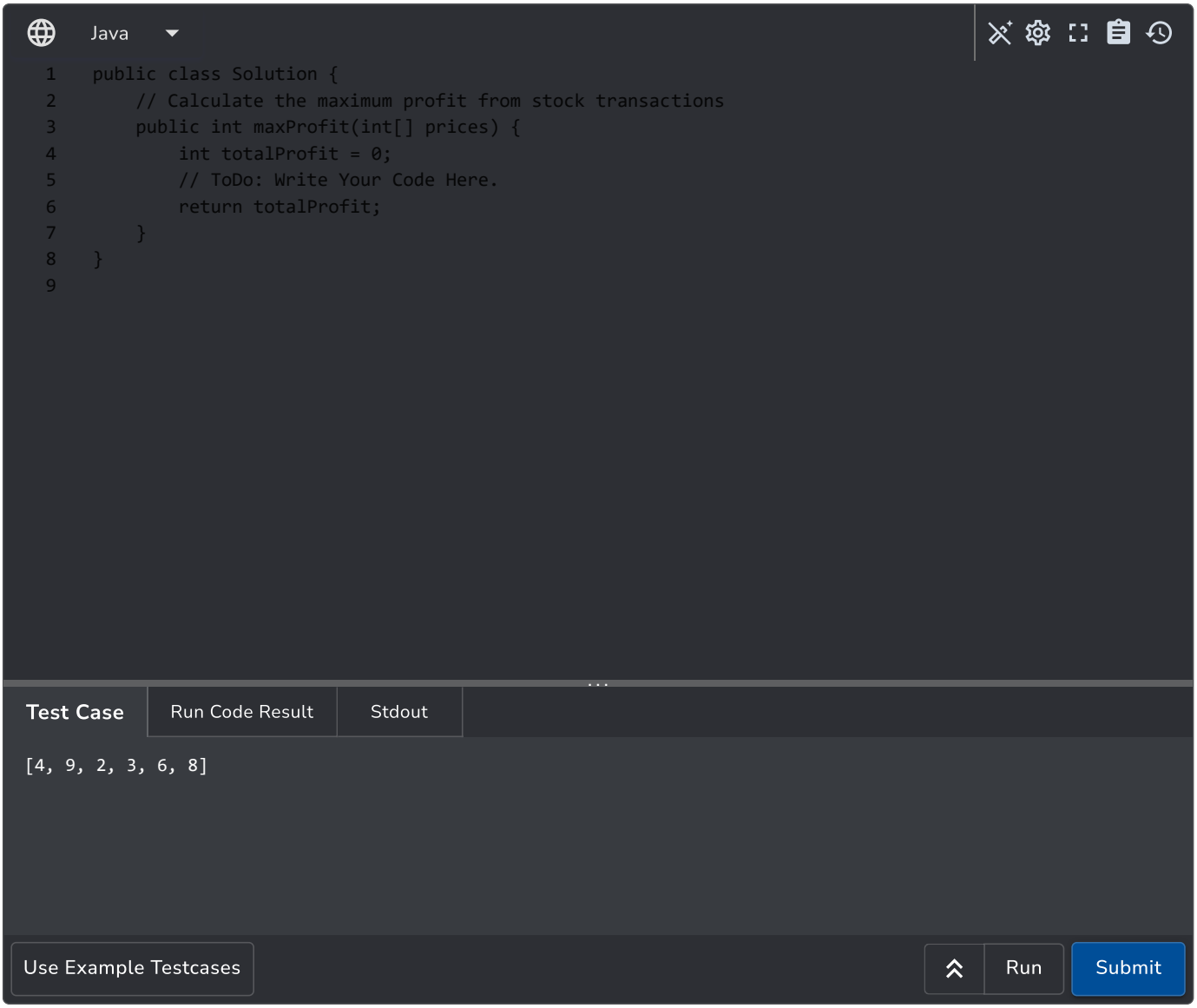
Find the `maximum profit` you can achieve from these transactions.

Examples

- **Example 1**:
  - **Input**: `[4, 9, 2, 3, 6, 8]`
  - **Expected Output**: `11`
  - **Justification**: Buy on day 1 (price = 4) and sell on day 2 (price = 9), profit = 9-4 = 5. Then buy on day 3 (price = 2) and sell on day 6 (price = 8), profit = 8-2 = 6. Total profit is 5 + 6 = 11.
- **Example 2**:
  - **Input**: `[5, 3, 6, 2, 5, 4]`
  - **Expected Output**: `6`
  - **Justification**: Buy on day 2 (price = 3) and sell on day 3 (price = 6), profit = 6-3 = 3. Then buy on day 4 (price = 2) and sell on day 5 (price = 5), profit = 5-2 = 3. Total profit is 3 + 3 = 6.
- **Example 3**:
  - **Input**: `[10, 8, 6, 5, 4, 2]`
  - **Expected Output**: `0`
  - **Justification**: Prices are decreasing each day, so there's no opportunity to make a profit. It is best not to make any transactions.

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Solution: Best Time To Buy And Sell Stock II

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